

Energy in Quantum physics and Quantum Chemistry

$$E = K + U$$

The total energy function in quantum physics and quantum chemistry is called the Hamiltonian (it is actually an operator rather than a function).

$$H = K + U$$

$$= \sum \frac{1}{2} m v^2 + U \quad (\text{multiple particles})$$

$$= \sum \frac{m^2 v^2}{2m} + U$$

$$= \sum \frac{p^2}{2m} + U$$

linear momentum

$$= \sum \frac{p^2}{2m} + \sum_i \sum_j \frac{k q_i q_j}{r_{ij}}$$

$$= \sum \frac{-\hbar^2}{2m} \frac{\Delta^2}{\Delta x^2} + \sum_i \sum_j \frac{k q_i q_j}{r_{ij}}$$

$$H\psi = E\psi \quad (\text{one form of Schrödinger wave equation})$$

ψ = the wave function

$\psi(r, t)$ gives the probability of finding a particle at position r at time t .

References

Engd, Thomas (2013). Quantum Chemistry & Spectroscopy (3rd ed). Pearson